Residual spectral graph learning for connectivity-loss estimation under multi-edge disruptions

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Abstract

Rapid evaluation of many simultaneous road-link disruptions requires a useful compromise between exact spectral recomputation and local approximation. We estimate the relative loss of algebraic connectivity after multi-edge deletion using a graph convolutional network (GCN) that learns a bounded correction to a first-order Fiedler sensitivity. The design is tested under independent and spatially clustered failures, with graph-disjoint synthetic splits, zero-shot transfer to five OpenStreetMap (OSM) networks, and geographically blocked leave-one-area-out transfer. Eight neural configurations and analytical or set-based baselines isolate the contribution of the residual prior, Fiedler node feature, coordinates, and message passing. All uncertainty is calculated from five seed-level replicates rather than treating correlated scenarios as independent. In synthetic-to-OSM zero-shot tests, residual learning reduces mean absolute error relative to a direct GCN by 0.0085 (95% seed-bootstrap interval 0.0039–0.0131) for independent failures and 0.0121 (0.0026–0.0247) for spatial failures. However, when models are trained on three OSM areas and tested on a held-out fourth, the direct GCN is more stable: its errors are 0.135 and 0.109, compared with 0.161 and 0.144 for the residual model. A reliability-context extension likewise improves synthetic performance but degrades OSM transfer. Controlled sparse-eigensolver scaling from 50 to 800 nodes separates one-time spectral setup, amortized cost, and update-only cost. These results identify the spectral residual as a useful but domain-sensitive inductive bias, and delimit its applicability to structural connectivity rather than hazard or traffic-flow prediction. Code and cached networks are available at github.com/lee-vtruong/ResiliRoad.

Keywords: algebraic connectivity; Fiedler vector; graph neural network; spatial network; infrastructure robustness; domain shift.

1 Introduction

Road networks are spatially embedded complex networks whose link losses can fragment access routes and alter system-wide connectivity. Exhaustive exact evaluation becomes costly when a planner must screen many multi-link scenarios. Conversely, a first-order perturbation is fast and interpretable, but its local linearity can fail under finite deletions, eigenvalue crossings, and disconnection. This tension motivates a hybrid estimator: retain the known spectral direction and learn only its nonlinear error.

Algebraic connectivity, the second-smallest Laplacian eigenvalue, is a global structural quantity [1, 2]. It does not measure travel demand, congestion, capacity, or recovery. The narrower goal here is fast structural stress testing: given an intact weighted graph and a set of removed edges, estimate its relative algebraic-connectivity loss. This distinction is important because transport robustness has also been defined using system travel time, capacity loss, accessibility, and repair trajectories [4, 5, 11].

This study asks four questions. **RQ1**: Does learning a residual over a Fiedler approximation improve direct graph regression? **RQ2**: Does this advantage persist under correlated disruptions and geographic domain shift? **RQ3**: Which gains come from the explicit prior, Fiedler node feature, coordinates, or adjacency-aware message passing? **RQ4**: Where does the estimator fail and how does its runtime scale?

The contributions are: (i) a bounded residual spectral–GCN for simultaneous edge deletions; (ii) controlled feature, architecture, and reliability-context ablations; (iii) two disruption processes on five cached OSM networks; (iv) both synthetic-to-real zero-shot and geographically disjoint OSM transfer; (v) seed-paired bootstrap inference, calibration and failure-regime diagnostics; and (vi) a benchmark separating eigensolver setup, amortized spectral evaluation, and update-only cost. Importantly, the transfer experiments reveal a limitation: the residual prior is not uniformly superior once OSM training data are available.

2 Related work

2.1 Spectral robustness of complex networks

Fiedler introduced algebraic connectivity as a graph invariant [1]; later work related Laplacian spectra to expansion, connectivity, synchronization, and robustness [2, 3]. For a simple eigenvalue, the squared difference of Fiedler coordinates across an edge gives the derivative with respect to its weight. Such derivatives explain infinitesimal sensitivity, whereas the present task includes several finite edge deletions. The residual model specifically targets the missing interaction and higher-order terms rather than replacing spectral analysis.

2.2 Road-network disruption

Road robustness depends on graph representation and disruption process. Multi-granularity analyses of empirical city networks show that structural conclusions can change with representation [6]. Link-based indices using flow or capacity address functional effects [4, 5], while critical-scenario optimization searches for damaging link combinations [7]. Hazard-independent studies compare random, localized, and targeted multi-link failures [10]; temporal analysis of Zurich also demonstrates process-dependent robustness [9]. Random road graph models enable controlled topology experiments [8]. Our spatial-cluster mechanism belongs to this stress-testing tradition but is a geometric proxy, not an empirical hazard model.

2.3 Graph learning and analytical priors

GCNs aggregate local neighbourhood information and are permutation equivariant at node level [12]. Deep Sets provides an invariant alternative that does not propagate along edges [13]. Graph Networks have also been used as learnable physical simulators [14], illustrating how known structure and learned corrections can coexist. The novelty claimed here is not the first use of GNNs, residual learning, or spectral metrics. It is the leakage-resistant empirical separation of an explicit spectral prior from a spectral node feature, followed by geographic transfer and negative-ablation analysis on spatial infrastructure networks.

3 Problem formulation and method

Let $G = (V, E, w)$ be a connected undirected weighted graph, with combinatorial Laplacian $L = D - A$ and eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots$. For disrupted edges $S \subseteq E$, the target is

$$y(G, S) = \operatorname{clip}\left(1 - \frac{\lambda_2(G - S)}{\lambda_2(G)}, 0, 1\right). \tag{1}$$

Thus a disconnected damaged graph has $y = 1$. If u_2 is a unit Fiedler vector and λ_2 is simple, edge sensitivity is

$$\frac{\partial \lambda_2}{\partial w_{ij}} = (u_{2,i} - u_{2,j})^2. \tag{2}$$

The analytical prediction sums intact-graph derivatives:

$$\widehat{y}_{\text{spec}} = \text{clip}\left(\frac{\sum_{(i,j)\in S} w_{ij}(u_{2,i} - u_{2,j})^2}{\lambda_2(G)}, 0, 1 \right). \tag{3}$$

Each damaged graph supplies normalized adjacency and node features: intact degree, failed-edge incidence, absolute normalized Fiedler coordinate, and two normalized coordinates. Three 48-unit GCN layers feed mean–max graph pooling and a two-layer head. The direct model predicts y; the residual model predicts

$$\widehat{y}_{\text{res}} = \text{clip}\left(\widehat{y}_{\text{spec}} + \tanh r_\theta(G, S), 0, 1\right). \tag{4}$$

The bounded correction concentrates capacity on finite-deletion error but can also inherit domain-specific bias from the prior.

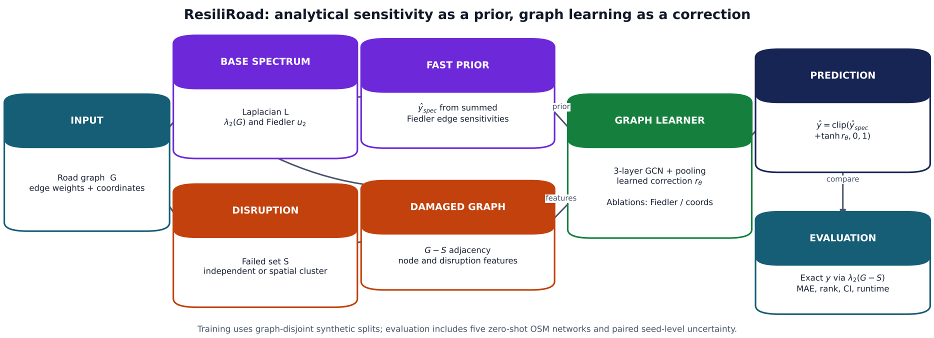

Figure 1: ResiliRoad workflow. Exact damaged-graph eigendecomposition constructs labels only. The intact Fiedler vector produces both an optional node feature and the explicit first-order prior; the GCN learns a bounded correction.
Alt text: A flow diagram splits an intact road graph into an exact-label branch and a fast spectral-prior branch. Damaged adjacency and node features enter a GCN whose residual is added to the spectral estimate.

The ablation matrix contains direct and residual GCNs with all features, without Fiedler, and without coordinates; Deep Sets and a summary-statistic MLP are non-GCN baselines. A further reliability-context residual appends the prior, normalized failure count, density, and graph size to the prediction head. It does not use damaged connectivity or any label-derived flag.

4 Experimental design

4.1 Synthetic graphs and disruption processes

Connected random geometric graphs contain 35–65 nodes and inverse-length edge conductance. Per seed, 800 scenarios are generated for each failure mode and entire base-graph families are assigned to train, validation, or test. Each scenario removes one to eight edges. Independent failure samples edges uniformly. Spatial-cluster failure chooses a random epicentre and removes the required number of edges with nearest geometric midpoints. Both modes share the same failure-count distribution.

4.2 OpenStreetMap networks

OSMnx [15] was used to obtain 650-m drivable networks around five Vietnamese sites. Directed multigraphs were simplified, converted to undirected simple graphs, and restricted to the largest connected component. Conductance is $100/\mathrm{length}_{\mathrm{m}}$. Cached GraphML freezes the exact instances because live OSM data evolve.

Table 1: OpenStreetMap networks used in the study.

Site	Nodes	Edges	Density
HCMUS, Ho Chi Minh City	163	221	0.0167
VIASM, Hanoi	206	301	0.0143
Da Nang	142	203	0.0203
Can Tho	114	178	0.0276
Da Lat	48	55	0.0488

Two transfer protocols answer different questions. In *zero-shot OSM*, all learned models train only on synthetic graphs and test on 100 scenarios per site, mode, and seed. In *leave-one-area-out OSM*, one site is test, the next site in a fixed rotation is validation, and the remaining three sites are training. Forty scenarios per site and mode are generated for each of five seeds; the direct and residual GCNs train for at most 25 epochs. No base area is shared across train, validation, and test in a fold.

4.3 Statistics, calibration, and runtime

Seeds 11, 22, 33, 44, and 55 are independent analysis units. MAE, RMSE, R^2, and Spearman correlation are first computed within seed. Confidence intervals resample the five seed-level statistics 10,000 times; comparisons

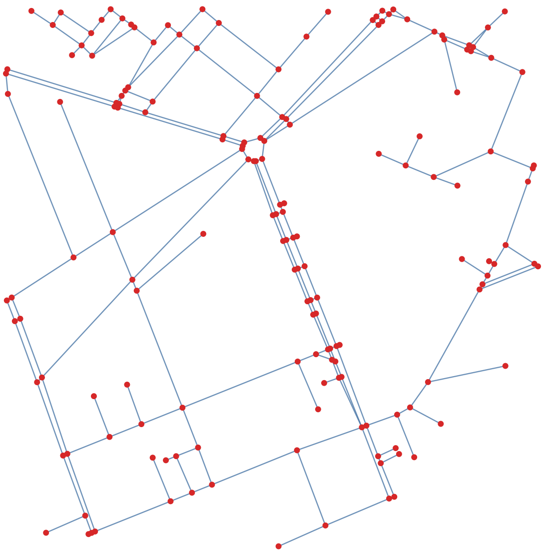

Figure 2: Processed HCMUS base network (163 nodes, 221 edges). Disruption scenarios remove subsets of the displayed links. Map data copyright OpenStreetMap contributors.

Alt text: A sparse, irregular street graph is drawn with red intersection nodes and pale-blue road edges; a dense central corridor connects several branching neighbourhoods.

bootstrap within-seed paired differences. No scenario-level p-value is used. Continuous calibration is summarized by regressing target on prediction within each seed, reporting slope, intercept, and signed mean error. Error slices by connectivity, failure count, density/site, and spectral clipping are diagnostic.

The original OSM benchmark uses dense symmetric eigendecomposition and one-scenario neural inference on CPU. A separate controlled benchmark uses sparse symmetric eigensolves on random geometric graphs with 50–800 nodes. It reports exact damaged-graph solution, update-only Fiedler sensitivity, and amortized spectral cost (one intact setup spread over ten scenarios). These denser graphs are computational stress tests rather than road-realistic samples.

5 Results

5.1 Graph-disjoint synthetic and zero-shot OSM results

Table 2: MAE as seed mean [95% seed-bootstrap interval].

	Synthetic		Zero-shot OSM	
Method	Independent	Spatial	Independent	Spatial
Spectral	.069 [.048,.095]	.144 [.102,.183]	.502 [.482,.521]	.650 [.643,.657]
Summary MLP	.065 [.051,.079]	.108 [.082,.135]	.281 [.240,.320]	.347 [.269,.425]
Deep Sets	.068 [.058,.078]	.105 [.085,.126]	.114 [.105,.123]	.143 [.122,.164]
Direct GCN	.077 [.055,.098]	.081 [.067,.097]	.106 [.100,.112]	.102 [.077,.148]
Residual GCN	**.052 [.036,.070]**	**.065 [.049,.079]**	**.097 [.091,.102]**	**.090 [.071,.124]**

The residual model improves direct-GCN MAE by paired differences 0.0253 [0.0106, 0.0400] and 0.0160 [0.0048, 0.0241] on synthetic independent and spatial failures. The improvements remain positive in zero-shot OSM: 0.0085 [0.0039, 0.0131] and 0.0121 [0.0026, 0.0247]. Yet the analytical prior alone has poor OSM calibration despite useful rank information. Removing Fiedler or coordinates from residual-full yields intervals containing zero; individual node-feature necessity is therefore not established.

5.2 Geographically blocked transfer

Table 3: Leave-one-area-out OSM performance, seed mean [95% seed-bootstrap interval]. Each fold uses three training, one validation, and one test area.

Method	Independent MAE	Spatial-cluster MAE
Direct GCN	**.135 [.123,.145]**	**.109 [.106,.112]**
Residual GCN	.161 [.118,.216]	.144 [.102,.198]

The point-estimate ranking reverses when training is moved into the OSM domain. Direct GCN has mean R^2 0.714 and 0.747, compared with 0.550 and 0.529 for the residual model. The paired residual-minus-direct MAE differences are 0.0261 [-0.0122, 0.0824] and 0.0350 [-0.0093, 0.0900]; both intervals include zero, so five seeds do not establish direct-model superiority. Residual performance is seed-sensitive: independent-failure MAE ranges from 0.092 to 0.265. Calibration explains part of this instability. Direct-GCN slopes stay between 0.91 and 1.06; residual slopes fall to 0.58 and 0.69 in the two poorest seeds, with substantial underprediction. Thus the spectral prior is helpful under one training distribution but can become a shortcut or a biased anchor under another.

5.3 Reliability-context negative ablation

Adding graph size, density, failure count, and spectral estimate to the residual head reduces synthetic spatial MAE by 0.0193 [0.0120, 0.0266] and has an inconclusive -0.0034 [-0.0159, 0.0111] change for synthetic independent failure (negative means improvement). It increases zero-shot OSM spatial MAE by 0.0304 [0.0063, 0.0583]; the independent increase is 0.0452 [-0.0029, 0.1019]. Simple reliability metadata therefore encourages distribution-specific calibration rather than robust transfer.

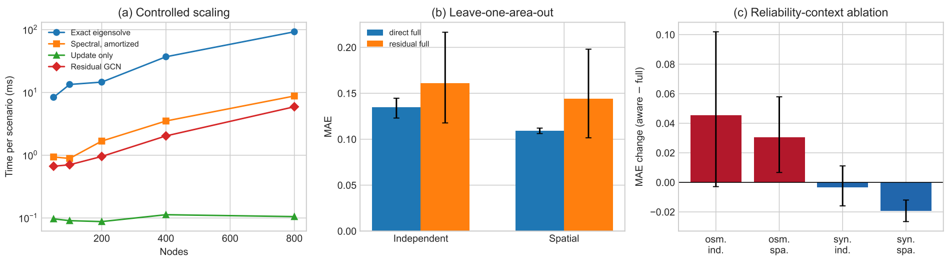

Figure 3: Journal extensions. (a) Controlled sparse-solver scaling separates exact, amortized, and update-only time. (b) Leave-one-area-out MAE. (c) MAE change caused by reliability context; positive values favour residual-full. Error bars in (b,c) are 95% seed-bootstrap intervals.

Alt text: Three panels show logarithmic runtime increasing with graph size, lower leave-one-area-out bars for direct GCN, and reliability context improving synthetic spatial data but worsening both OSM conditions.

5.4 Runtime and error regimes

On the five small OSM networks, exact dense eigendecomposition averages 5.094 ms per scenario, first-order update 0.018 ms, and residual inference 0.674 ms on CPU. Controlled sparse scaling gives exact times 8.37, 13.41, 14.62, 37.03, and 92.61 ms for 50, 100, 200, 400, and 800 nodes. Update-only time remains 0.088–0.113 ms; amortized spectral cost is 0.94, 0.89, 1.68, 3.51, and 8.79 ms, while residual-GCN inference is 0.67, 0.70, 0.96, 2.03, and 5.92 ms. The slight non-monotonicity at small size reflects fixed solver overhead and graph-to-graph variability; this benchmark is descriptive, not a complexity proof.

In zero-shot OSM, residual-full MAE is 0.070 for connected damaged graphs and 0.112 after disconnection. Site errors range from 0.041 (Can Tho) to 0.189 (Da Lat). The densest site is also the smallest and hardest, so density cannot be interpreted causally. No evaluated first-order estimate clips at one; the requested clipped/unclipped comparison is therefore undefined. The largest leave-one-area-out errors are retained in a machine-readable table instead of being removed as outliers.

6 Discussion

The central positive result is conditional. A spectral residual improves a direct GCN on graph-disjoint synthetic tests and when both are transferred zero-shot to OSM. It combines the prior's ranking signal with learned nonlinear calibration. However, blocked OSM transfer and reliability-context ablation show that this inductive bias can amplify domain-specific error. Direct prediction is more stable when the training distribution already contains real-road graphs. This boundary condition is scientifically more useful than a universal superiority claim.

The study also separates the explicit prior from Fiedler node information. Neither Fiedler nor coordinate removal produces a stable loss, whereas replacing the residual architecture does. The robust ingredient in the first protocol is therefore how mathematical and learned estimates are combined, not merely the presence of an eigenvector channel. The poor Deep Sets ranking in several regimes supports adjacency-aware propagation, while the summary MLP shows that coarse graph statistics are insufficient.

6.1 Limitations

Five 650-m Vietnamese neighbourhoods do not establish city-scale or cross-country generality. Their size and density are confounded. The synthetic geometric generator is denser than the OSM graphs, and the scaling graphs are deliberately denser still. Spatial clusters are not calibrated to flood, landslide, earthquake, construction, or conflict data. Algebraic connectivity is structural: it omits direction, demand, congestion, capacity, travel time, accessibility, and restoration. Only five seed replicates limit precision, and no probabilistic predictive intervals are produced. Sparse batched neural implementations may change runtime rankings at city scale.

Future work should pre-register larger metropolitan regions, use nested blocked validation across countries, and factor graph size from density. Hazard rasters could define correlated failures, while traffic assignment could test whether structural loss predicts service degradation. Mixture-of-experts or learned gating may decide when to trust the prior; conformal methods could add graph-population prediction regions. These extensions should preserve area-level replication rather than inflate sample size with correlated scenarios.

7 Conclusion

Residual spectral graph learning offers an interpretable, fast estimator of finite multi-edge connectivity loss, but its advantage depends on training domain. The method beats direct regression under graph-disjoint synthetic and synthetic-to-OSM zero-shot evaluation; direct GCN is more stable under blocked OSM-to-OSM transfer. Reporting both outcomes, along with ablations, hierarchical uncertainty, calibration, error cases, and separated runtime costs, provides a reproducible account of when a Fiedler prior helps and when it does not.

Data and code availability

Code, cached OSM-derived GraphML, manifests, experiment scripts, environment versions, and compact result tables are available at `https://github.com/lee-vtruong/ResiliRoad`. OSM data are licensed under the Open Data Commons Open Database License and are attributed to OpenStreetMap contributors [16]. Before submission, the author will archive the exact release in a DOI-issuing repository and replace this sentence with its permanent data citation.

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Conflict of interest

The author declares no conflict of interest.

Acknowledgements and AI-use disclosure

Generative AI tools were used for language editing and software-development assistance. The author designed the study, verified the implementation and results, and takes responsibility for the manuscript.

A Reproducibility details

All models use AdamW, mean-squared error, validation early stopping, and one PyTorch compute thread. Core runs use at most 80 epochs; blocked OSM transfer uses at most 25. Software versions are Python 3.13.11, NumPy 2.4.4, SciPy 1.17.1, NetworkX 3.6.1, pandas 2.2.3, scikit-learn 1.9.0, PyTorch 2.11.0 (CPU), Matplotlib 3.11.1, OSMnx 2.1.1, and GeoPandas 1.1.4. The machine has 16 logical AMD64 CPUs; CUDA was unavailable. Scripts record exact seeds and emit scenario predictions, training histories, fold assignments, runtime summaries, bootstrap tables, calibration diagnostics, and worst cases.

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